



BAULKHAM HILLS HIGH SCHOOL

2015 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks) **Questions 1-10**

- Attempt Questions 1-10
Allow about 15 minutes for this section.

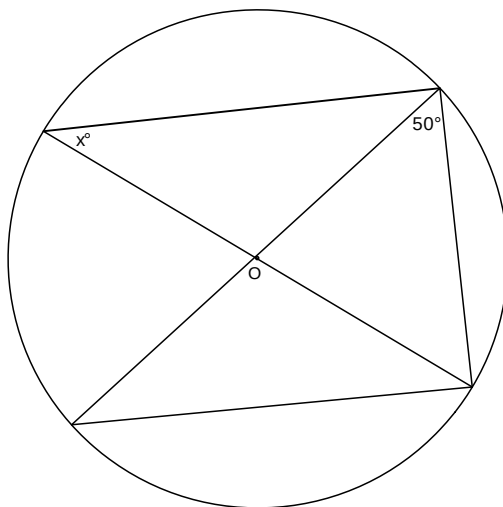
Section II – Pages 5-10 (60 marks)

- Attempt questions 11-14
Allow about 1 hour and 45 minutes for this section.

Table of Standard Integrals is on page 11

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. If O is the centre of the circle, the value of x in the following diagram is:

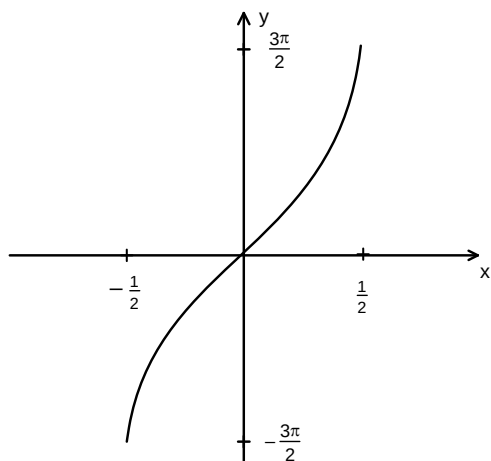


- (A) 25°
- (B) 40°
- (C) 50°
- (D) 80°
2. The point P divides the interval AB externally in the ratio 3 : 2. If A(-2,2) and B(8,-3) what is the y coordinate of the point P?
- (A) -13
- (B) -1
- (C) 4
- (D) 28
3. How many distinct arrangements of the letters of the word *ALGEBRA* are possible in a straight line if the A's are separated.
- (A) 720
- (B) 1800
- (C) 2160
- (D) 2520

4. The polynomial $P(x) = x^3 - 6x^2 - 2x + k$ has a factor of $x + 2$. What is the value of k ?
- (A) -28
(B) -20
(C) 20
(D) 28

5. The acute angle between the lines $4x + y = 2$ and $y = 2x - 1$ to the nearest degree is :
- (A) 12°
(B) 13°
(C) 40°
(D) 41°

6. The equation of an inverse trig function drawn below is :



- (A) $y = \frac{1}{3} \sin^{-1} \frac{x}{2}$
(B) $y = \frac{1}{3} \sin^{-1} 2x$
(C) $y = 3 \sin^{-1} \frac{x}{2}$
(D) $y = 3 \sin^{-1} 2x$
7. A particle moving in simple harmonic motion with displacement x and velocity v , is given by $v^2 = 9(16 - x^2)$. What is its amplitude (A) and its period (T) ?
- (A) $A=3$ $T = \frac{\pi}{2}$
(B) $A=3$ $T = \frac{2\pi}{3}$
(C) $A=4$ $T = \frac{\pi}{2}$
(D) $A=4$ $T = \frac{2\pi}{3}$

8. $\int \frac{1}{\sqrt{25-4x^2}} dx =$

(A) $\frac{1}{2} \sin^{-1} \frac{2x}{5} + c$

(B) $\frac{1}{2} \sin^{-1} \frac{4x}{25} + c$

(C) $\frac{1}{4} \sin^{-1} \frac{2x}{5} + c$

(D) $\frac{1}{4} \sin^{-1} \frac{4x}{25} + c$

9. The derivative of $\tan^{-1}x^4$ is:

(A) $\frac{1}{1+x^8}$

(B) $\frac{4x^3}{1+x^8}$

(C) $4(\tan^{-1}x)^3$

(D) $\frac{4(\tan^{-1}x)^3}{1+x^2}$

10. The solution to $|2x-1| \leq |x-2|$ is

(A) $x \leq -1$

(B) $x \geq 1$

(C) $-1 \leq x \leq 1$

(D) $x \leq -1$ or $x \geq 1$

End of Section 1

Section II – Extended Response**All necessary working should be shown in every question.**

Question 11 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Solve $\frac{2}{3x-1} \leq 1$	3
b)	Find $\int \frac{2x \, dx}{(2x+1)^2}$ using the substitution $u = 2x + 1$	3
c)	Evaluate $\lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{3x}$	2
d)	Find the constant term in the expansion $\left(2x + \frac{3}{x^3}\right)^8$.	2
e)	(i) Show that a root of the continuous function $f(x) = x + \frac{1}{2}\sin 2x - \frac{\pi}{4}$ lies between 0.4 and 0.5.	1
	(ii) Hence use one application of Newton's method with an initial estimate of $x = 0.4$ to find a closer approximation for the root to 2 significant figures.	2
f)	Solve $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$	2
End of Question 11		

Question 12 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Evaluate $\int_0^{\frac{\pi}{8}} \sin^2 2x \, dx$	3
b)	(i) From a group of 6 boys and 6 girls, 8 are chosen at random to form a group. How many different groups of 8 people can be formed? (ii) How many of these groups consist of 4 boys and 4 girls? (iii) 4 boys and 4 girls are chosen and placed around a circle. What is the probability that the boys and girls alternate?	1 1 2
c)	The rate of change of the temperature (T) of an object is proportional to the difference between the temperature of the object and the temperature of the surrounding medium(C), ie $\frac{dT}{dt} = k(T - C)$ An object is heated and placed in a room of temperature 20°C to cool. After 10 minutes its temperature is 36°C. After 20 minutes the temperature is 30°C. (i) Show $T = C + Ae^{kt}$ is a solution to the differential equation above. (ii) Find the value of A and the value of k to 3 decimal places. (iii) What was the temperature of the object when it was first placed in the room?	1 3 1
d)	Prove $(1)(3)(5) \dots (2n+1) = \frac{(2n+1)!}{2^n n!} \text{ for } n \geq 0 \text{ by mathematical induction.}$	3
End of Question 12		

Question 13 (15 marks) - Start on the appropriate page in your answer booklet**Marks**

a) The polynomial $P(x) = x^3 - 6x^2 + kx - 8$ has roots α, β and γ .

Find :

(i) $\alpha + \beta + \gamma$

1

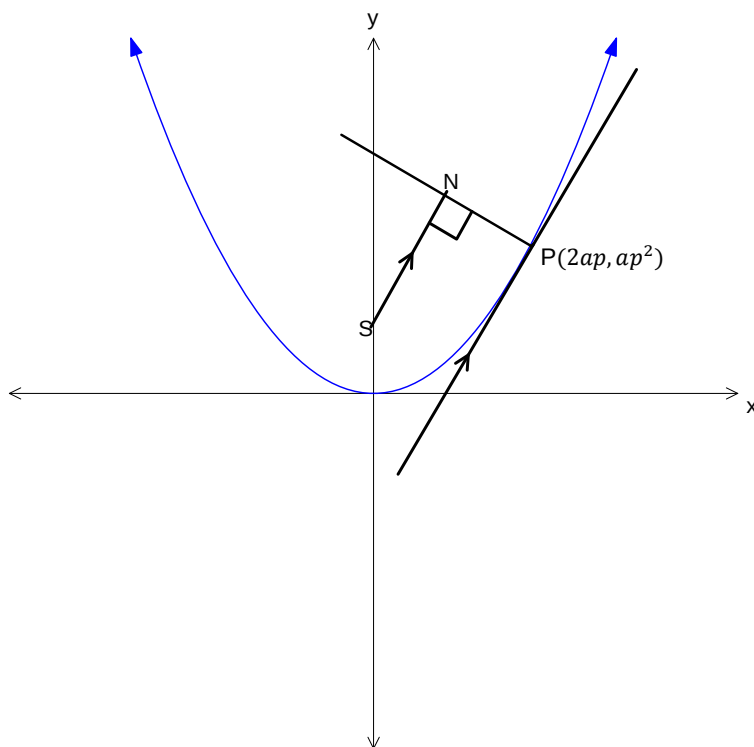
(ii) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

2

(iii) k if $P(x)$ has a triple root.

2

b) PN is the normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$. The normal intersects the line SN which is parallel to the tangent at P . S is the focus of the parabola.



(i) Show the equation of the normal PN is $x + py = 2ap + ap^3$.

2

(ii) Find the equation of the line SN .

1

(iii) Show that N has coordinates $(ap, ap^2 + a)$.

2

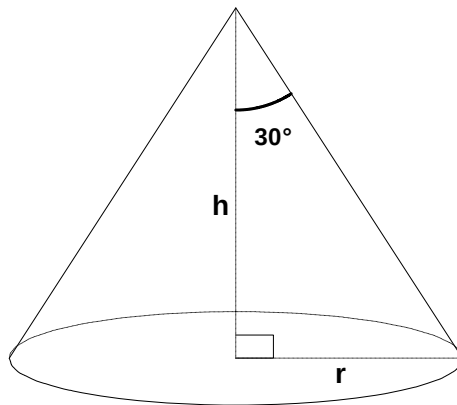
(iv) Find the equation of the locus of N as P moves on the parabola.

2

Question 13 continues on the following page

c) Sand is falling on the ground forming a conical pile whose semi apex angle is 30° .

The volume of the pile is increasing at a rate of $\frac{\pi}{100} m^3/s$. ($V = \frac{1}{3}\pi r^2 h$)



(i) Show that the volume of the pile is given by: $V = \frac{\pi h^3}{9}$

1

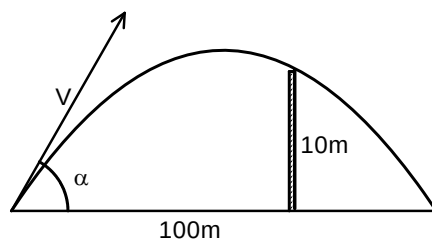
(ii) Find the rate at which the height of the pile is increasing when the height of the pile is 2 metres.

2

End of Question 13

Question 14 (15 marks) - Start on the appropriate page in your answer booklet

a)

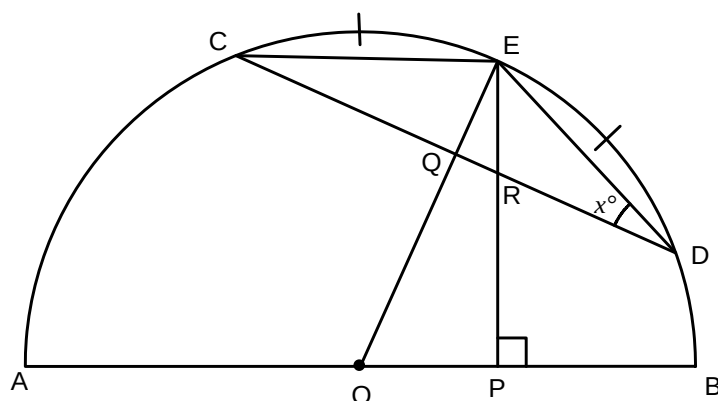


A projectile is fired from the ground with an angle of projection given by $\alpha = \tan^{-1} \frac{3}{4}$ and initial velocity V .

It just clears a wall $10m$ high $100m$ away. Let acceleration due to gravity be $g=10ms^{-2}$.

- | | | |
|-------|---|---|
| (i) | Show that the equations of motion are $x = \frac{4Vt}{5}$ and $y = -5t^2 + \frac{3Vt}{5}$. | 2 |
| (ii) | Find the initial velocity, V of the projectile. | 2 |
| (iii) | At what speed is the projectile travelling the instant it clears the wall? | 2 |

b) Copy or trace the diagram below in your exam booklet.

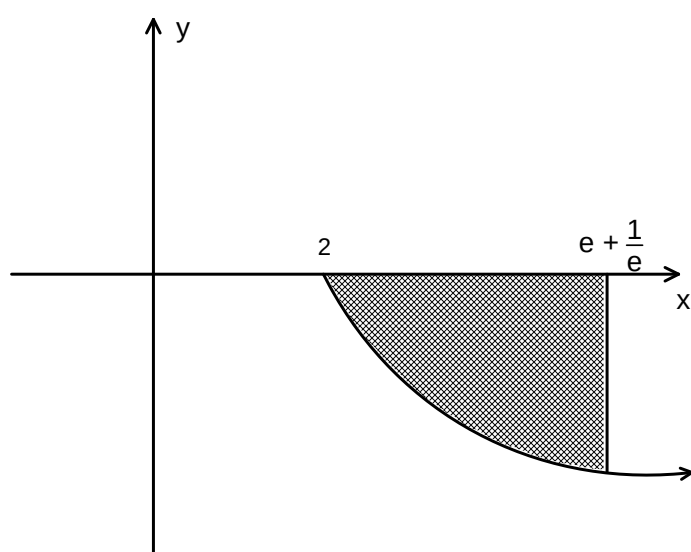


E is the midpoint of the arc CD . The radius OE meets CD at Q . EP is perpendicular to the diameter AB and meets CD at R .

- | | | |
|------|--|---|
| (i) | If $\angle CDE = x^\circ$, show $\angle EOD = 2x^\circ$. | 2 |
| (ii) | Prove $OPRQ$ is a cyclic quadrilateral. | 2 |

Question 14 continues on the following page

c) Below is the graph of $y = \ln \left(\frac{x - \sqrt{x^2 - 4}}{2} \right)$



- (i) Show that the equation of the inverse function is given by $y = e^x + e^{-x}$. 2
- (ii) Hence find the area of the shaded region above. 3

End of Paper.

M. Choice

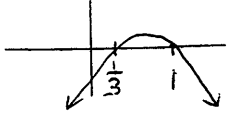
- 1) B 2) A 3) B 4) D 5) D
6) D 7) D 8) A 9) B 10) C

Question 11.

a) $\frac{2}{3x-1} \leq 1 \quad x \neq \frac{1}{3}$

$$\frac{2(3x-1)^2 - 1(3x-1)^2}{(3x-1)^2} \leq 0 \quad \checkmark$$

$$\frac{(3x-1)(2-(3x-1))}{(3x-1)(3-3x)} \leq 0 \quad \checkmark$$



b) $\therefore \text{Soln } x < \frac{1}{3} \text{ or } x > 1$

$$\int \frac{2x}{(2x+1)^2} dx \quad u = 2x+1 \quad \frac{du}{dx} = 2$$

$$\therefore \int \frac{2x}{(2x+1)^2} dx \quad dx = \frac{du}{2}$$

$$= \int \frac{u-1}{u^2} \cdot \frac{du}{2}$$

$$= \frac{1}{2} \int \left(\frac{u}{u^2} - \frac{1}{u^2} \right) du$$

2015 EXT.1. TRIAL SOLUTIONS: -

$$= \frac{1}{2} \int \left(\frac{1}{u} - u^{-2} \right) du \quad \checkmark$$

$$= \frac{1}{2} \left(\ln u + \frac{1}{u} \right) + C$$

$$= \frac{1}{2} \left(\ln(2x+1) + \frac{1}{2x+1} \right) + C \quad \checkmark$$

c) $\lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{3x}$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{\frac{1}{2}x} \cdot \frac{\frac{1}{2}x}{3x} \quad \checkmark$$

$$= 1 \times \frac{1}{3} \quad \checkmark$$

$$= \frac{1}{6}$$

d) $(2x + \frac{3}{x^3})^8$

$$\text{General term} = {}^8C_k (2x)^{8-k} \left(\frac{3}{x^3}\right)^k$$

$$= {}^8C_k 2^{8-k} x^{8-k} \cdot 3^k x^{-3k}$$

$$= {}^8C_k 2^{8-k} 3^k x^{8-4k} \quad \checkmark$$

$$\therefore 8-4k = 0$$

$$k = 2$$

$$\therefore \text{Constant term} = {}^8C_2 2^6 3^2 = 16128 \quad \checkmark$$

e) (i) $f(0.4) = -0.0267 \dots$

$$f(0.5) = 0.135 \dots$$

since $f(0.4) < 0$ & $f(0.5) > 0 \quad \checkmark$
a root exists between 0.4 & 0.5

(ii) $f'(x) = 1 + \cos 2x$

$$\therefore a_1 = 0.4 - \left(\frac{-0.0267}{1.696} \right)$$

$$= 0.4157 \dots \quad \checkmark$$

$$a_1 = 0.42 \quad \checkmark$$

f) $\sin 2\theta = \cos \theta$

$$2 \sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (2 \sin \theta - 1) = 0$$

$$\sin \theta = \frac{1}{2} \quad \cos \theta = 0$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2} \quad \checkmark$$

Question 12

2) $\int_0^{\frac{\pi}{8}} \sin^2 2x \, dx$

$$= \frac{1}{2} \int_0^{\frac{\pi}{8}} (1 - \cos 4x) \, dx \quad \checkmark$$

$$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{8}} \quad \checkmark$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{8} - \frac{\sin \frac{\pi}{2}}{4} \right) - (0 - 0) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{8} - \frac{1}{4} \right]$$

$$= \frac{\pi}{16} - \frac{1}{8} \text{ or } \frac{\pi-2}{16} \quad \checkmark$$

$$= (0.0713 \dots)$$

b) (i) ${}_{12}C_8 = 495 \quad \checkmark$

(ii) ${}_6C_4 \times {}_6C_4 = 225 \quad \checkmark$

(iii) $\frac{\text{Alternating}}{\text{Total}} = \frac{4! \times 3!}{7!} \quad \checkmark$

$$= \frac{1}{35} \quad \checkmark$$

(or equivalent)

c) (i) $T = C + Ae^{kt} \rightarrow Ae^{kt} = T - C \quad \text{--- (1)}$

$$\frac{dT}{dt} = kAe^{kt} \text{ from (1) } \quad \checkmark$$

$$= k(T - C)$$

$$\therefore T = C + Ae^{kt} \text{ is a soln}$$

Given when $t = 10$ $T = 36$ $C = 20$

$$\therefore 36 = 20 + Ae^{10k}$$

$$Ae^{10k} = 16 \quad \text{--- (1)}$$

when $t = 20$ $T = 30$

$$\therefore 30 = 20 + Ae^{20k}$$

$$\therefore 10 = Ae^{20k} \quad \text{--- (2)}$$

(2) $\div$ (1) $\frac{10}{16} = \frac{Ae^{20k}}{Ae^{10k}}$

$$\frac{5}{8} = e^{10k}$$

$$\therefore k = \frac{\ln(\frac{5}{8})}{10} = (-0.047) \quad \checkmark$$

Sub into (2) $10 = Ae^{-0.047(20)}$

$$A = 25.6 \quad \checkmark$$

$$T = 20 + 25.6e^{kt}$$

when $t = 0$

$$T = 20 + 25.6e^0$$

$$= 45.6 \quad \checkmark$$

d) $(1)(3)(5) \dots (2n+1) = \frac{(2n+1)!}{2^n n!} \quad n \geq 0$

Step 1.

Prove true for $n = 0$

$$\text{LHS} = 1 \quad \text{RHS} = \frac{(2(0)+1)!}{2^0 0!} = \frac{1!}{1} = 1$$

Step 2: LHS = RHS (true for $n = 0$)

Assume true for $n = k$ i.e.

$$(1)(3)(5) \dots (2k+1) = \frac{(2k+1)!}{2^k k!}$$

Step 3

Prove true for $n = k+1$ i.e.

$$(1)(3) \dots (2k+1)(2k+3) = \frac{(2k+3)!}{2^{k+1} (k+1)!} \quad \checkmark$$

$$\text{i.e. } \frac{(2k+1)!}{2^k k!} \cdot (2k+3) = \frac{(2k+3)!}{2^{k+1} (k+1)!}$$

$$= \frac{(2k+1)! (2k+2)(2k+3)}{2^k k! (2k+2)}$$

$$= \frac{(2k+3)!}{2^{k+1} k! 2(k+1)} \quad \checkmark$$

$$= \frac{(2k+3)!}{2^{k+1} (k+1)!}$$

$$= \text{RHS.}$$

Step 4: True for $n = k+1$ if true for $n = k$. Therefore true for $n = 0, 1, 2, \dots$ by M. Induction.

$$(14c) \quad y = \ln\left(\frac{x - \sqrt{x^2 - 4}}{2}\right)$$

inverse
✓ $x = \ln\left(\frac{y - \sqrt{y^2 - 4}}{2}\right)$

$$e^x = \frac{y - \sqrt{y^2 - 4}}{2}$$

$$2e^x = y - \sqrt{y^2 - 4}$$

$$\sqrt{y^2 - 4} = y - 2e^x \quad \checkmark$$

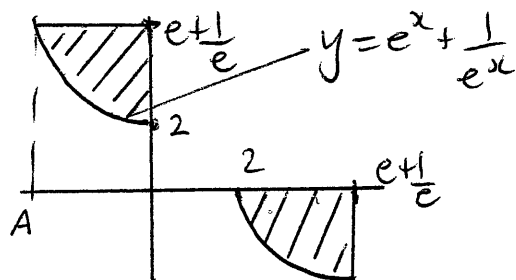
$$y^2 - 4 = y^2 - 4ye^x + 4e^{2x}$$

$$4ye^x = 4e^{2x} + 4$$

$$y = \frac{4e^{2x}}{4e^x} + \frac{4}{4e^x}$$

$$y = e^x + \frac{1}{e^x}$$

$$\underline{y = e^x + e^{-x} \quad \checkmark}$$



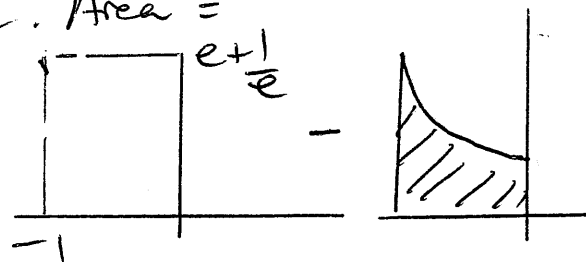
2 areas are equal because of symmetry.

At A $y = e + \frac{1}{e}$

$$\therefore e + \frac{1}{e} = e^x + \frac{1}{e^x}$$

✓ $x = \pm 1$ by inspection.
here $x = -1$

$\therefore$ Area =



$$\begin{aligned} &= \left(e + \frac{1}{e}\right) \times 1 - \int_{-1}^0 e^x + e^{-x} \quad \checkmark \\ &= e + \frac{1}{e} - (e^x - e^{-x})_{-1}^0 \\ &= e + \frac{1}{e} - (e^0 - e^0 - (e^{-1} - e^1)) \\ &= e + \frac{1}{e} - (1 - 1 - \frac{1}{e} + e) \end{aligned}$$

$$\begin{aligned} &= e + \frac{1}{e} + \frac{1}{e} - e \\ &= \frac{2}{e} \quad \checkmark \end{aligned}$$

NB AWARD 2 for

$$\int_{-1}^0 e^x + e^{-x} = e - \frac{1}{e}$$

AWARD 2

$$\int_0^1 e^x + e^{-x} = e - \frac{1}{e}$$